The Promised Trillion Dollar AI-Coding Market

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Abstract

Forecasts say AI coding will be worth trillions of dollars a year. This essay lists who says so, labels each number by what it is, and names the assumptions the forecasts share.

In plain words

1. The big numbers about AI coding are mostly guesses about the future, not counts of money already earned.
2. Almost all of them assume that every new piece of code needs a large AI model, paid by the word, forever.
3. If working code can be saved and reused, the cost of a repeated job falls toward the cost of the electricity to run it, and those forecasts would be too high.

1. Projections and measurements are different objects

Every number below carries one of five labels.

Label	Meaning
Projection	A forecast or a value estimate built on stated assumptions
Company-reported	A figure a company states about itself, such as annualized run-rate revenue (the latest month times twelve) or a funding valuation
Survey estimate	A market-sizing figure from a survey of buyers
Filed	A figure in a regulatory filing
Measured	A count or an energy reading from a run whose artifacts are kept. Energy readings carry one of three labels: `est_j` for an analytical estimate, `reported_j` for an operating system or on-chip counter, `measured_j` for an external power meter

Only the last row is evidence about the cost of a correct result. The others are evidence about what the market expects.

2. Who projects what

2.1 Developer value: the trillion

Claim	Number	Label	Source
Software developers worldwide	about 30 million (27 million to 47 million by estimate)	Projection input	Appenzeller and Li [1]
Value per developer per year	$100,000 (assumed)	Projection input	[1]
Annual economic contribution	about $3 trillion	Projection	[1]
Developers on GitHub	180 million plus; 518.7 million pull requests merged in the year	Company-reported	GitHub Octoverse 2025 [17]
Value of generative AI across 63 use cases	$2.6 trillion to $4.4 trillion a year; software engineering productivity 20% to 45% of current annual function spend	Projection	McKinsey [15]
Global GDP effect of generative AI	7% over a ten-year period	Projection	Goldman Sachs [16]
Productivity gain from a simple assistant	about 20%	Estimate from interviews	[1]
Productivity gain from a best-of-breed deployment	“at least double”	Estimate “based on anecdotal evidence”	[1]

The developer count is soft: 30 million is a professional estimate, and GitHub alone counts six times that many accounts that build on it [17]. The per-developer value and the productivity multiple are soft too (Section 2.7). McKinsey’s 2023 estimate is the original trillion of this market, and it is a value estimate across all functions [15]. The same venture essay says the AI coding ecosystem can “support dozens of billion-dollar companies and even a trillion-dollar giant” [1]. That sentence is the trillion. It is a value estimate. It is not a revenue forecast.

2.2 Tool revenue and enterprise spend

Claim	Number	Label	Source
Enterprise spend on AI coding tools, 2025	$4.0 billion, 55% of departmental AI spend	Survey estimate	Menlo Ventures [2]
Enterprise generative AI spend, 2025	$37 billion	Survey estimate	[2]
AI code tools market, 2026	$9.35 billion, to $29.96 billion by 2031	Projection	Mordor Intelligence [3]

The revenue the market reports today is in the billions. The trillion is the value the market hopes to capture.

2.3 Company revenue and valuations

Company claim	Number	Date	Label	Source
Cursor annualized revenue	over $1 billion	13 November 2025	Company-reported	Cursor [5]
Cursor Pro repriced	from request limits to “$20 of frontier model usage per month at API pricing”, with refunds for the change	June to July 2025	Company-reported	Cursor [18]
Cursor Series D	$2.3 billion at $29.3 billion post-money	13 November 2025	Company-reported	[5]
Anthropic run-rate revenue	$14 billion	12 February 2026	Company-reported	Anthropic [6]
Claude Code run-rate revenue	over $2.5 billion	12 February 2026	Company-reported	[6]
Anthropic Series G	$30 billion at $380 billion post-money	12 February 2026	Company-reported	[6]

Anthropic states that the round funds “infrastructure expansions” [6]. Revenue, valuation and compute are one bet.

Figure 1 sets the dollar figures of Sections 2.1 to 2.3 and 2.6 on one axis, marked by the label each source carries.

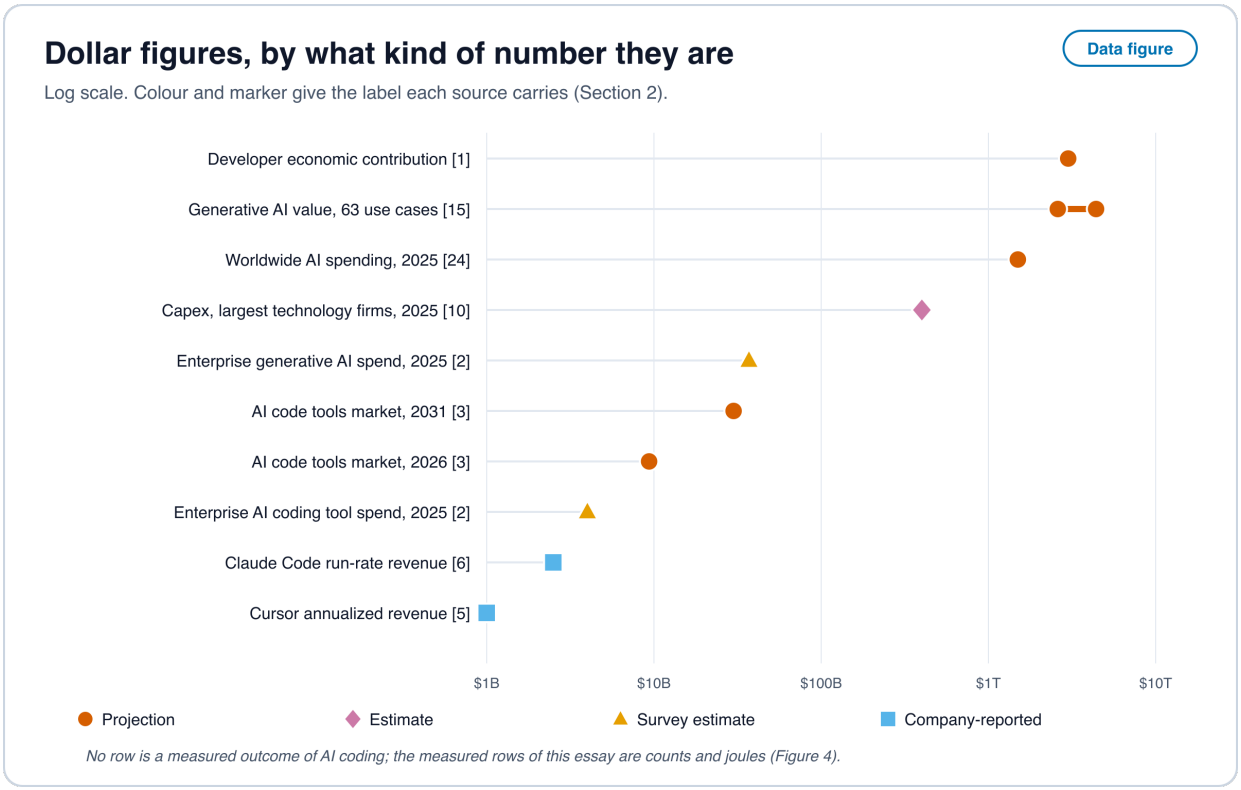


Figure 1. Dollar figures by label. Data figure, log scale. Sources: Appenzeller and Li [1], Menlo Ventures [2], Mordor Intelligence [3], Cursor [5], Anthropic [6], IEA [10], McKinsey [15], Gartner [24]. A PNG copy is provided.

2.4 Inference, tokens and unit economics

Claim	Number	Label	Source
Google tokens processed per month	9.7 trillion (2024), about 480 trillion (2025), over 3.2 quadrillion (May 2026)	Company-reported	Pichai [8]
Google internal AI developer tools	half a trillion tokens a day in March 2026, more than three trillion a day in May	Company-reported	[8]
Cost of one agentic coding query at list prices	$2.50: a 100,000-token context and 10,000 output tokens at $15 and $75 per million tokens	Illustration	Appenzeller and Li [1]
The same at 3 queries an hour, 7 hours a day, 200 days	about $10,000 a year per developer	Illustration	[1]
Tokens in agentic coding against code reasoning and code chat	about 1,000 times more; runs of one task differ by up to 30 times; higher token use “does not translate into higher accuracy”	Measured (eight models, SWE-bench Verified)	Bai, Huang and Wang [9]
Net losses from Microsoft’s investment in OpenAI, quarter ended 30 September 2025	$4.1 billion pre-tax, its equity-method share, in other expense; a $3.1 billion reduction in net income ($0.41 per diluted share)	Filed and company-reported	Microsoft Form 10-Q [7]; Microsoft FY26 Q1 performance [19]

Two lines in the sources state the economics plainly. Pichai: “many companies are already blowing through their annual token budgets, and it’s only May” [8]. Appenzeller and Li: “For decades, software development cost was almost purely people’s cost, but now LLMs add a substantial opex component” [1].

Figure 2 plots the token series.

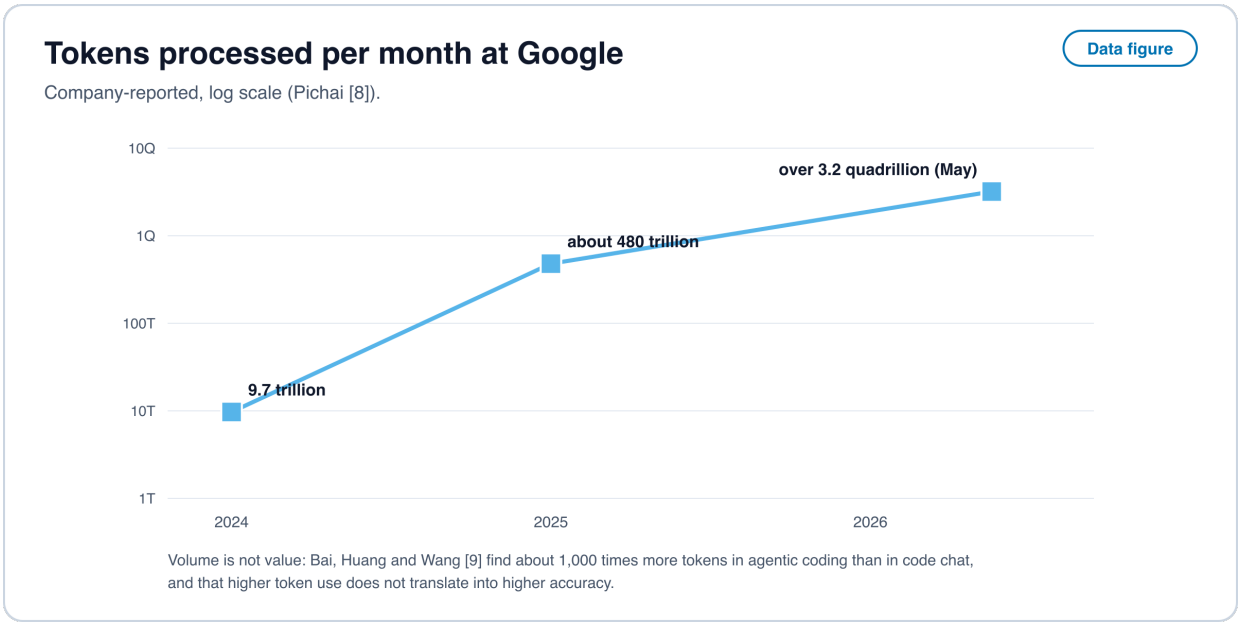


Figure 2. Tokens processed per month at Google. Data figure, company-reported, log scale. Source: Pichai [8]; the note on token use is from Bai, Huang and Wang [9]. A PNG copy is provided.

2.5 Price per token

Claim	Number	Label	Source
Cost of a fixed level of AI performance	down about 47% per quarter, about 13 times per year, since 2023	Measured price series (five benchmarks)	Epoch AI [4]
Price to reach a fixed benchmark score	falls 9 to 900 times a year, depending on the milestone; fastest drops most recent and least certain to persist	Measured price series (log-linear fits)	Epoch AI [34]
Inference cost at equivalent performance	down about 10 times a year: $60 per million tokens in 2021 to $0.06	Estimate	Appenzeller, a16z [20]
DeepSeek API price cut, 29 September 2025	developers' cost "lowered by more than 50%"	Company-reported	DeepSeek [21]

The 13 times a year of [4] is one point in a wide range [34]. The fall is driven by competition and open weights, as the DeepSeek cut shows [21]. It needs no energy accounting. That is the first of the two declines in Section 5. Reuse adds a different one: calls per task.

Figure 3 draws the two price series.

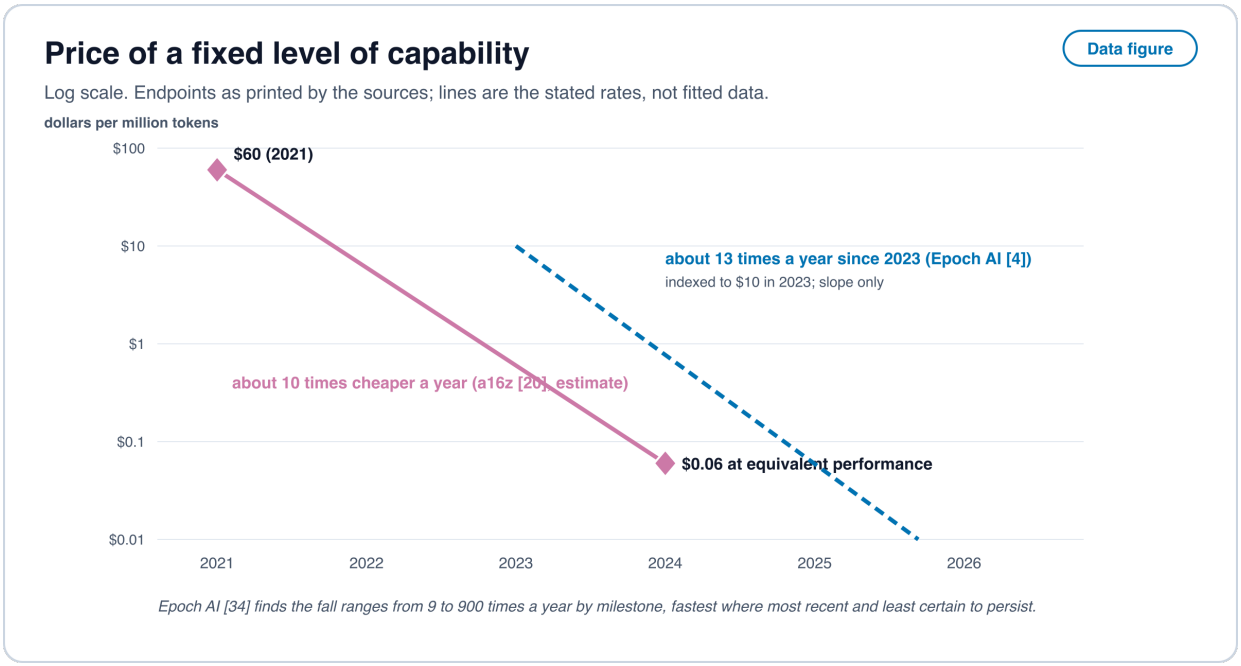


Figure 3. Price of a fixed level of capability. Data figure, log scale: the a16z endpoints are as printed [20]; the Epoch AI line is the stated rate of about 13 times a year since 2023 [4], indexed to an arbitrary start to show slope only. The range by milestone is from Epoch AI [34]. A PNG copy is provided.

2.6 Energy and capital

Claim	Number	Label	Source
Data-centre electricity, 2024	about 415 TWh, about 1.5% of world electricity	Estimate	IEA, *Energy and AI* [11]
Data-centre electricity, 2030	about 945 TWh, Base Case	Projection	[11]
Data-centre electricity, 2025 to 2030	485 TWh to 950 TWh, about 3% of world demand	Estimate and projection	IEA, *Key Questions on Energy and AI* [10]
Capex of the largest technology companies	over $400 billion in 2025, expected up another 75% in 2026	Reported and projected	[10]
Alphabet capex	$31 billion in 2022; about $180 to $190 billion expected in 2026	Company-reported and guidance	Pichai [8]
Energy per task	down by at least an order of magnitude a year; reasoning and agentic tasks use “hundreds or thousands of times more energy per query than simple text generation”	Estimate	[10]
One median text prompt to a production assistant	0.24 Wh	Measured by the operator	Elsworth et al. [12]
Stargate	$500 billion over four years for AI infrastructure for one lab, $100 billion deployed at once	Company-announced	OpenAI [22]
Worldwide AI spending, 2025	about $1.5 trillion	Projection	Gartner [24]
Revenue the capex must earn back	“AI’s $600B question”	Analysis	Sequoia Capital [23]

The rent that must be serviced is hardware rent. Capex and commitments of this size are paid back through usage priced above its energy. On tasks that can be closed by reuse, that base is smaller.

2.7 Productivity evidence

Study	n	Design	Effect	Source
a16z essay	interviews	Estimate	about 20% for a simple assistant; "at least double" best of breed	[1]
GitHub Copilot, HTTP server task	recruited developers	Controlled experiment	55.8% faster on one toy task	Peng et al. [27]
Google enterprise trial	96 engineers	Randomized controlled trial	about 21% less time on task	Paradis et al. [28]
METR, early 2025	16 experienced open-source developers, 246 tasks	Randomized controlled trial	19% slower with AI	Becker et al. [25]
METR, 2026 update	new cohort	Randomized controlled trial	the authors call the new data "an unreliable signal"	METR [26]

The measured effects range from a 19% slowdown to a 56% speed-up on a toy task. Both factors of the value estimate, the developer count and the productivity multiple, are soft. That weakens the trillion. It also means that any value freed by cheaper coding is not a known number.

3. What the projections assume

Read together, the projections rest on four assumptions. Each is a statement about one direction in the design space, not about the value of software.

1. One direction: generation. Every change is written by a model. The value estimate scales productivity by the model, and the cost illustration prices every query [1].
2. High energy and high compute. Token volume, data-centre electricity and capex are projected to grow together [8, 10, 11]. The energy per query falls, and the tokens per task rise faster [9, 10].
3. Per-token rent. Revenue is billed per token or per seat whose cost is tokens. A developer's annual token cost is projected at the scale of a salary [1].
4. AI in the loop forever. Task 10,000 costs what task 1 costs, because nothing the model learned on task 1 runs without the model on task 10,000.

The essay that states the trillion also states its limit: "while substitution pricing works in early markets, eventually the cost of a good converges to its marginal cost, and so does pricing" [1]. This essay agrees, and asks what the marginal cost of a correct coding task is.

4. What a correct task costs

Define, for one coding task, $C \in \{0, 1\}$ the outcome of its acceptance check (tests pass, a proof checks, or an output matches a reference), J the energy used to reach $C = 1$, in joules, c_J the price of energy, in dollars per joule, and c_{check} the cost of running the check, in dollars. A buyer pays for $C = 1$, not for tokens.

Energy bounds marginal cost. The marginal cost of a run is at least $c_J J$, since the electricity must be bought. Under competition, price tends toward marginal cost [1, 14].

Much code is reused. Gurevich proved that every sequential algorithm is captured by a sequential abstract state machine: a set of states and a transition rule [35]. Lopes and colleagues measured that about 70% of the code on GitHub consists of clones of previously created files [36]. Where a passing solution already exists, a task can be closed by retrieving and checking it rather than generating it again.

Proposition E0 (price of a covered task). Call a task covered when a stored solution passes its check. Let J_{floor} be the energy to retrieve, apply and check that solution. Under competition among sellers whose results pass the same check, the price of a covered task tends toward

$$p_{\text{covered}} \to c_J J_{\text{floor}} + c_{\text{check}}.$$

Generation is needed only on uncovered tasks. For a task stream with covered fraction $f \in [0, 1]$ and mean generation cost p_{gen} on uncovered tasks, the expected price per task is $\bar{p} = f\, p_{\text{covered}} + (1 - f)\, p_{\text{gen}}$. The fraction f is not yet

measured (Proposition E4).

5. Where the projections are fragile

Per-token pricing. A task closed by reuse needs no tokens, so a revenue line priced per token earns nothing on it. Total token revenue on a task stream scales with $1 - f$.

The model in every loop. If a solution found once is stored and checked, later tasks of the same shape do not need the model, so the model calls per task fall as f grows. Whether they do on real work is measured by Proposition E3.

Energy per task. One median text prompt to a production assistant is measured by its operator at 0.24 Wh, or 864 J [12], and the IEA puts reasoning and agentic queries at hundreds or thousands of times the energy of a simple one [10]. No public measurement compares generation and retrieval on the same tasks, so no ratio is stated here.

Two declines multiply. Write the cost per task as $p = n \cdot q$, with n model calls per task and q dollars per call. The price per call at fixed performance falls by a factor of 9 to 900 a year, depending on the task [4, 34]. Reuse lowers n. A projection that extrapolates token volume models only q.

Bundling. A good whose marginal cost is near zero, and which complements a tool people already have, tends to be bundled into that tool, and its separate price disappears [13, 14]. Nalebuff showed that a bundle can deter entry [32], and Adams and Yellen that it can discriminate on price [33]; Spolsky described the strategy of commoditizing a complement [31]. Pricing changes are visible: GitHub moved Copilot's individual plans to usage-based billing from 1 June 2026 [30], and Cursor moved Pro from request limits to pass-through API pricing in 2025 [18].

What survives. The value estimate of Section 2.1 can hold. What is fragile is the assumption that the value is collected as a charge on generation, since that charge applies only to the fraction $1 - f$.

6. Demand

Cheaper tasks can raise total demand. The venture essay expects it: "we believe Jevon's Paradox in this case holds" [1]. Two effects should be separated. Demand for one task stops once $C = 1$. Total use can still grow, because cheaper work opens new tasks. The one published elasticity is small. Across tasks in Anthropic's API sample, each 1% increase in cost is associated with a 0.29% decrease in usage frequency, an elasticity of $\varepsilon \approx -0.29$ [29]. That is cross-sectional across tasks, not a price change over time. With $|\varepsilon| < 1$, a price decline on a fixed task set lowers revenue, since revenue $R = pQ$ changes by $dR/R = (1 + \varepsilon)\, dp/p$.

7. Propositions and protocols

Proposition E1 (the cost curves cross). For a stream of tasks in one repository, the cumulative cost per resolved task of a method that stores and reuses checked solutions falls below that of a method that generates every change, after a finite number of tasks. *Protocol:* run both on the same issue stream under an external power meter. Report `measured_j` and dollars per resolved task by task index, and the crossing index.

Proposition E2 (price follows the floor). Where a coding capability can be closed by retrieval and rules, its list price falls toward $c_J J_{\text{floor}} + c_{\text{check}}$. *Protocol:* track list prices of such a capability class against its measured run energy over time.

Proposition E3 (the token share of cost). In a repository where reuse grows, the share of task cost paid as tokens falls. *Protocol:* log tokens, joules and dollars per closed task by round, and report the token share of cost by round.

Proposition E4 (coverage). The covered fraction f of real coding task volume is measurable. *Protocol:* draw a random sample of merged pull requests from public repositories (GitHub reports 518.7 million merged in a year [17]), test each for a retrieved solution that passes the repository's tests, and report f with a 95% Wilson interval.

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